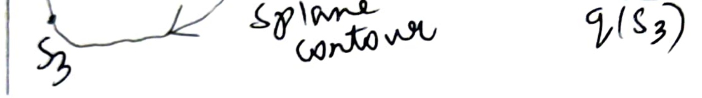
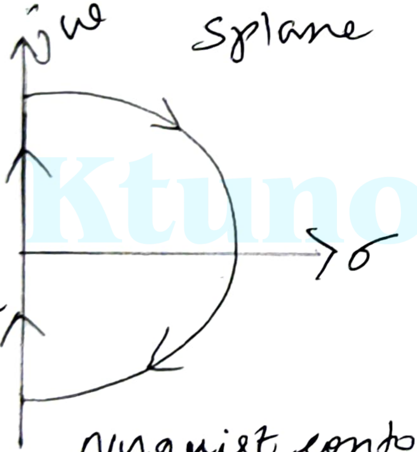
graphically from OL freq respons
Nyquist stability criterion is
a theorem of complex variables
principle of arguments.
consider a function $q(s)$

$$q(s) = \frac{(s-\alpha_1)(s-\alpha_2)\cdots(s-\alpha}{(s-\beta_1)(s-\beta_2)\cdots(s-}$$

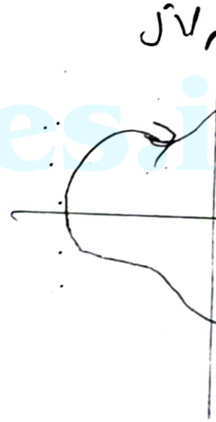
if we put 's' a value we will get q
's' be a complex variable $s=$
complex 's' plane ∴ the functio
is also a complex (depends on
& may be defined as $q(s) = u$
For every point 's' in the s plan
which $q(s)$ is analytic we ca
a corresponding $q(s)$ in the q
For a contour in the 's' plan
doesnot go through any sin
there correspond a contour
$q(s)$ plane [A function $q(s)$ i
in the splane provided the fu
all its derivatives exist ie f

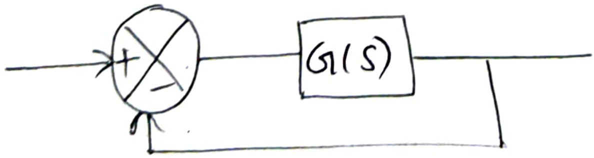


Stable - no poles on right side so we consider this contour covering entire right side



Nyquist contour





$$G(S) = \frac{(S+Z_1)(S+Z_2)}{(S+P_1)(S+P_2)} \quad \text{—— (1)}$$

$$CE \Rightarrow 1+G(S)=0$$

S^2 term, ∴ so Z_1, Z_2

$$\frac{(S+P_1)(S+P_2)+(S+Z_1)(S+Z_2)}{(S+P_1)(S+P_2)}$$

③ & ④ ⇒ zeros of CE = CL p

For CL s/m to be stable the
(ie CL poles) should not be lo
the right half of s plane.
Evenif some OL poles (ie CE po
the right half of s plane all
(ie CE zeros) should lie in
half s plane for a stable

Nyquist contour

It is a contour w
completely encloses the right h
splane. It helps to find the
of any CE zero (CL poles) in
half of s plane.
It is directed clockwise & c
of an infinite line seg
along the $j\omega$ axis & an arc C_2
radius
Along C_1 $s = j\omega$ where s varies
$-j\infty$
Along C_2 $s = Re^{j\theta}$ with θ varyi
$R \to \infty$
$\frac{\pi}{2}$ to

The nyquist contour ... enclose all the right half s pl... zeros & poles of $q(s) = 1+G(s$...

let there be 'Z' zeros (CLP) roots of CE) ... 'P' poles (OLP) of q(s) (CE) in Re... half of s plane (nyq contour). A ... along the nyquist contour in ... the closed contour q(s) enclose ... origin of q(s) contour N ... in the counter clockwise directi...

$N = P - Z$

For the CL s/m to be stable th... should not be any zeros of q(... in the right half s plane (nyq ...

ie $Z = 0$

$N \rightarrow$ No. of anticlockwise encirc... of q(s) contour w.r.to origin ... no. of anticlockwise encircle... G(s)H(s) contour w.r.to -1

$Z \rightarrow$ zeros of q(s) (CE) located ... the s contour. (right half of ...

$P \rightarrow$ poles of CE lying on right ... s plane.

net encirclement

$G(s)H(s) = [1+G(s)H(s)] - 1$

$= q(s) - 1$

ie contour $q(s)$ corresponding nyquist contour in the s plane as contour $G(s)H(s)$. Encircle of origin $q(s)$ by $q(s)$ contour as encirclement of pt $-1+j$ $G(s)H(s)$ contour.

Nyquist stability criteria can be stated as if $G(s)H(s)$ contour corresponding to nyquist contour in the s plane encircle pt $-1+j0$ in the anticlockwise direction many times as the no: of right s plane poles of $G(s)H(s)$ the CL sm is stable

$G(s)H(s)$ contour

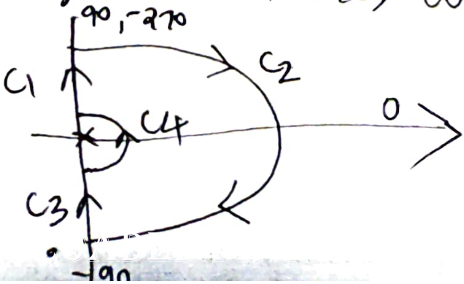
when $z=0$ $N=P$

3) clockwise encirclement of
In this case s/m is always u

Procedure for obtaining sta
using nyquist stability criter

) choose a nyquist contour wh
encloses the entire right h
except singular pt. singular
poles on imaginary axis

2) The nyquist contour
Should be mapped in
the G(s) H(s) plane to
determine the encirclement
of -1+j0 pt in the G(s)H(s) plan
The nyquist contour can be
into 4 sections C_1 C_2 C_3 C_4
Of these 4 sections can be d
sectionwise and then comb
together to get G(s) H(s) contou



$G(j\omega)H(j\omega)$. Equate the
to zero to find the phase
freq. substituting this freq
part the crossing pt of
on the real axis is obtain
the knowledge of type no: &
of the s/m approx locus of
can be plotted. ie value of
at $\omega=0$ & $\omega=\infty$

iii) separating mag & phase of G
equate phase to $-180°$ & solv
This ω is ω_{pc} & mag at
is crossing pt on real ax
approx polar plot as mention

4) C_2

The section C_2 of Nyquist
is a semicircle of infinit
hence the mapping of C_2 fro
to $G(s)H(s)$ plane can be obtai
putting $s = \lim_{R\to\infty} Re^{j\theta}$. θ v
from $+\pi/2$ to $-\pi/2$

$$G(s)H(s) = \frac{K1}{\lim_{R \to \infty} (Re^{j\theta})^{n-m}}$$

$$= 0\, e^{-j\theta(n-m)}$$

$\theta = \frac{\pi}{2}$ $G(s)H(s) = 0e^{-j\pi/2}$

$\theta = -\pi/2$ $G(s)H(s) = 0\, e^{j\pi}$

∴ section C_2 is mapped as ... or circular arc arou... with radius tending t...

5) C_3

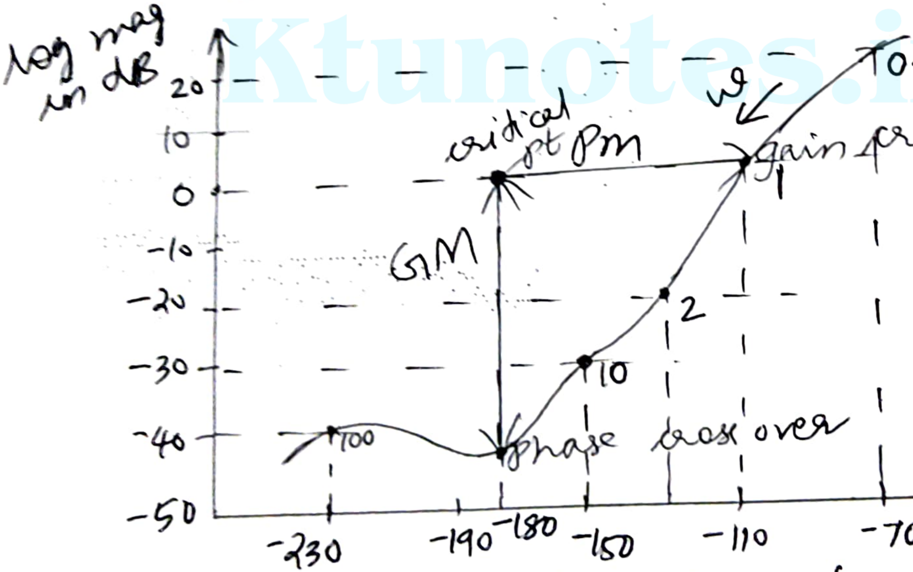
mapping of section ... obtained by putting $s = j$... $G(s)H(s)$ & w varied from ... This locus is the inverse ... of $G(j\omega)H(j\omega)$ ie mirro... w.r.to real axis

6) C_4 $G(s)H(s) = \frac{K(1+sT_1)(1+ ...}{s^y(1+sT_a) ...}$

$\theta = \pi/2 \quad G(s)H(s) = \infty e^{-j\pi/2}$

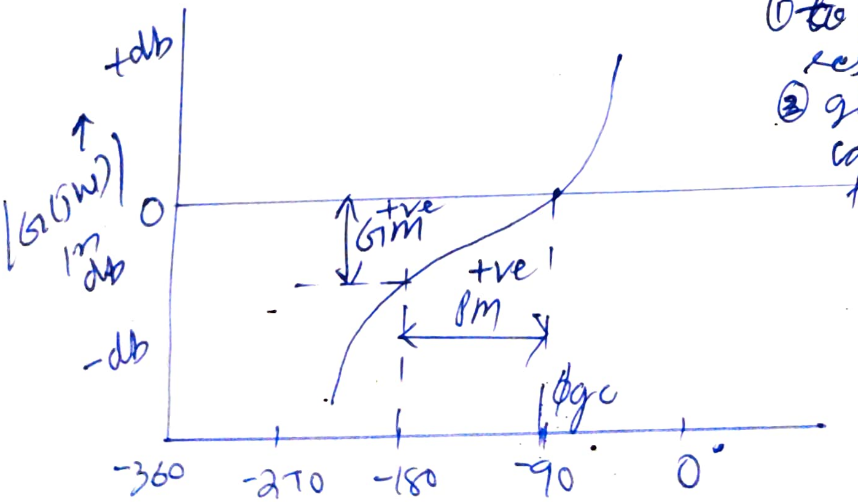
section C_4 is a semicircle
radius mapping of section
obtained by putting $s = \lim_{R \to \infty}$
$G(s)H(s)$ with θ varying from
$\therefore$ section C_4 is mapped as
arc with origin as centre
infinite radius

plot is constructed. The
are relative stability of CL
can be determined quickly
compensation can be carried
easily.



→ critical pt - intersection of 0dB
→ Phase cross over is where
→ gain cross over " 0dB axis
→ GM - vertical distance in dB measured
cross over to critic
→ PM - horizontal dist in degrees m
from gain cross over to

Nichols plot of OL s/m co
on the ordinary graph. Th
plot is a graph b/w ma
G(jw) in db along Y axis
of G(jw) along x axis for va
adv ② The frequency domain spe
can be determined from
usually choice of freq are co
Fix all points on ordinary
& Join points by smooth



① to
re
② g
co

Determine PM of uncomp s/m
Bode plot if PM doesnot
the requirement then lag comp

Step 4

choose a suitable value for
PM of comp s/m

$$\gamma_n = \gamma_d + \epsilon$$

γ_n = PM of comp s/m γ_d = des
ϵ - additional phase lag to
for shift in ω_{gc} choose
value of $\epsilon = 5^\circ$

Step 5

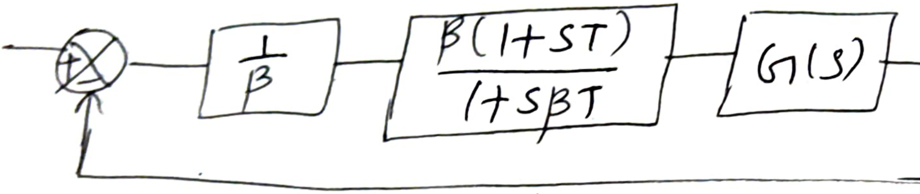
Determine new gain cross o
ω_{gcn}. ω_{gcn} is the freq
to γ_n on Bodeplot of uncom

Step 6

Determine β of compensato
β is given by mag of G
ω_{gcn} let the dB gain

$$A_{gcn} = 20 \log \beta.$$

Determine OLTF of



OLTF of comp s/m $G_0(s) = \frac{1}{\beta} \frac{\beta(1}{1+}$

$$= \frac{1+sT}{1+s\beta T} \cdot G(s)$$

Step 9

Determine the actual PM of
s/m $\gamma_0 = 180 + \phi_{gc0}$

ϕ_{gc0} = phase angle of $G_0(j\omega)$ at ω

if the actual PM satisfies
specification design is accept
otherwise repeat steps 4 to
taking $\varepsilon = 5^\circ$ more than pre

$\phi_m = \gamma_d + \varepsilon - \gamma$

ϕ_m - max phase lead angle

γ_d - desired PM

γ = PM of uncomp s/m.

ε - additional phase lead
compensate for shift in ω_g

if $\phi_m > 60$ then 2 cascade co
each having $\phi_{m\,new} = \frac{\phi_m}{2}$

initial value of $\varepsilon = 5°$

5) Determine TF of comp

6) calculate α using $\alpha = \frac{1-\sin}{1+\sin}$

from bode plot determine
the freq at which mag of G
$-20\log\frac{1}{\sqrt{\alpha}}$ This freq is ω_m

calculate T from the rela

$$\omega_m = \frac{1}{T\sqrt{\alpha}}$$

TF of lead comp

$$= \frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}} = \frac{\alpha(1+sT)}{1+s\alpha T}$$

Design of lag lead comp usi

lag lead comp is employed
large error const & large
required.

First design a lag section & so
& design a lead section

1) Determine OL gain k of un
s/m to satisfy specified error

2) Draw Bode plot of uncomp s/m

3) determine pm from Bodeplot

4) choose a new pm $\gamma_n =$

$\varepsilon = 5^\circ$

5) from bode plot determine the
gain cross over freq (ω_{gcn})
is the freq corresponding to
γ_n. choose the gain cross
of lag comp. ω_{gcl} some v
greater than ω_{gcn} $\therefore$ ω_{gc}

βT_1

TF of lag section

$$G_l(S) = \frac{S + \frac{1}{T_1}}{S + \frac{1}{\beta T_1}} = \beta \frac{(1+S\text{…}}{(1+S\beta\text{…}}$$

8) Determine TF of lead sec…

Take $\alpha = \frac{1}{\beta}$

From Bode plot find ω_m …
is the freq at which the …
is $-20\log\frac{1}{\sqrt{\alpha}}$

$$\omega_m = \frac{1}{T_2\sqrt{\alpha}}$$

$$T_2 = \frac{1}{\omega_m\sqrt{\alpha}}$$

TF of lead section

$$G_2(S) = \frac{S + \frac{1}{T_2}}{S + \frac{1}{\alpha T_2}} = \alpha \frac{(1+\text{…}}{(1+\text{…}}$$

or not.

Determine ω_{pc} of sm with

$$G(s) = \frac{1}{s(1+2s)(1+s)}$$

Ans

$$G(j\omega) = \frac{1}{j\omega(1+2j\omega)(1+j\omega)}$$

$$\angle G(j\omega) = -90 - \tan^{-1}2\omega - \tan^{-1}\omega$$

at ω_{pc} $\angle G(j\omega) = -180$

$$-180 = -90 - \tan^{-1}2\omega - \tan^{-1}\omega$$

taking tan on both sides $\tan[\tan^{-1}2\omega + \tan^{-1}\omega] = \infty$

$$\frac{2\omega + \omega}{1 - 2\omega^2} = \infty$$

$$1 - 2\omega^2 = 0$$

$$\omega^2 = \frac{1}{2}$$

$$\omega_{pc} = 0.70$$

of s/m.

→ from δ & ω_n time specifications can be

Q $G(s)H(s) = \dfrac{k}{s(s+4)(s+5)}$ find

s = −1 is on RL or n
angle criteria

<u>Ans</u> $G(s)H(s)\Big|_{s=-1} = \dfrac{k}{-1\cdot(3)(}$

<u>angle criterion</u>
$\angle G(s)H(s) = \pm 180(2q+1)$
$q = 0,1,2,3$
ie odd multiple of $180°$

$= \dfrac{k+j0}{(-1+j0)(3+j0)(}$

$= \dfrac{0}{180°\ 0°\ 0°}$ =

hence point s = −1 is

find value of k at w
is one of the root o

$\left|G(s)H(s)\right|_{s=-1} = 1$

$$\overline{\quad s^2+4s+13 \quad}$$

Ans
$$\frac{k(-4+2j+2)}{(-4+2j)^2+4(-4+2j)+1\ldots}$$

$$= \frac{k(-2+2j)}{(16-16j-4)+-16\ldots}$$

$$= \frac{k(-2+2j)}{9-8j}$$

$|G(j\omega)H(j\omega)| \neq 1$

$\angle G(j\omega)H(j\omega) \neq -18\ldots$

hence not a po…